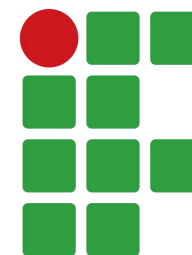


PROGRAMAÇÃO PARA DISPOSITIVOS MÓVEIS

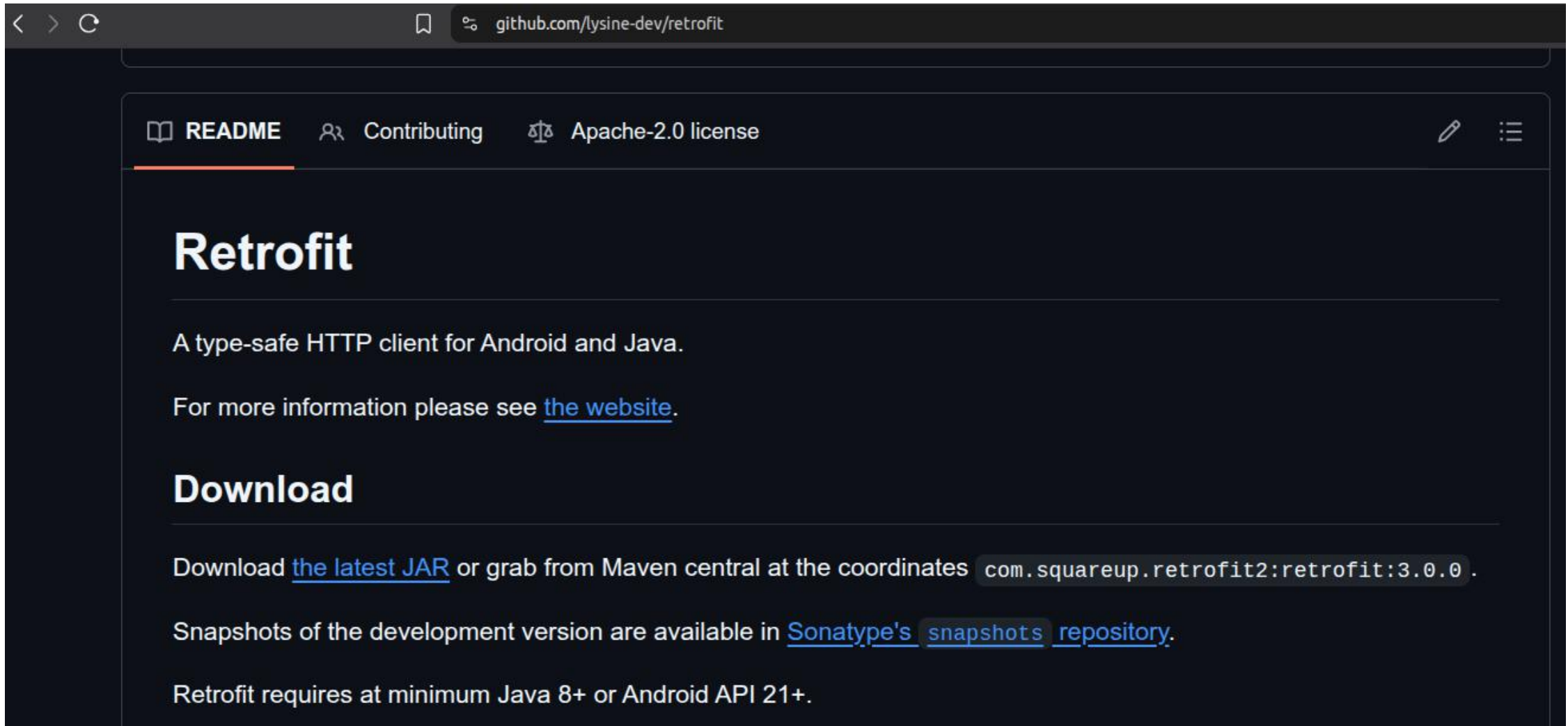
Curso de Engenharia de Software
Lucas Sampaio Leite



**INSTITUTO
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Retrofit

- Retrofit é uma biblioteca desenvolvida pela Square que facilita o consumo de APIs REST em aplicações Android e Java.
- Simplifica o envio e o recebimento de dados por meio de requisições HTTP a serviços REST.
- Permite trabalhar facilmente com dados no formato JSON, convertendo-os entre objetos Kotlin/Java e JSON.
- Permite utilizar diferentes conversores para a serialização e desserialização dos dados, como o Gson.



< > C github.com/lysine-dev/retrofit

README Contributing Apache-2.0 license

Retrofit

A type-safe HTTP client for Android and Java.

For more information please see [the website](#).

Download

Download [the latest JAR](#) or grab from Maven central at the coordinates `com.squareup.retrofit2:retrofit:3.0.0`.

Snapshots of the development version are available in [Sonatype's snapshots repository](#).

Retrofit requires at minimum Java 8+ or Android API 21+.

Retrofit

- O Retrofit não é exclusivo do Android, mas é fortemente associado ao ecossistema Android.
 - É uma biblioteca Java/Kotlin para consumo de APIs HTTP/REST.
 - Pode ser utilizada em outros projetos JVM, desde que sejam compatíveis com suas dependências.
 - No Android, tornou-se muito popular por simplificar a criação de clientes HTTP e a conversão de dados, especialmente JSON.

Retrofit

New Project

×

Empty Activity

Create a new empty activity with Jetpack Compose

Name

CurrencyConverterApplication

Package name

leite.sampaio.lucas.currencyconverterapplication

Save location

/home/lucas/AndroidStudioProjects/CurrencyConverterApplication

Minimum SDK

API 24 ("Nougat"; Android 7.0)

?

Your app will run on approximately 99,2% of devices.

Help me choose

Build configuration language ?

Kotlin DSL (build.gradle.kts) [Recommended]

Previous

Next

Cancel

Finish

Configurando dependências manualmente

- gradle/libs.versions.toml

```
[versions]
retrofit = "3.0.0"
converter-gson = "3.0.0"
```

```
[libraries]
retrofit = { group = "com.squareup.retrofit2", name = "retrofit", version.ref = "retrofit" }
retrofit-gson = { group = "com.squareup.retrofit2", name = "converter-gson", version.ref = "converter-gson" }
```

- build.gradle.kts (Module)

```
dependencies {
    implementation(libs.retrofit)
    implementation(libs.retrofit.gson)
```

Permissões

- No Android, as permissões (permissions) são mecanismos de controle que regulam o acesso dos aplicativos a dados, recursos e funcionalidades sensíveis do dispositivo.
- Elas permitem restringir o acesso dos aplicativos a determinados recursos, como localização, câmera, microfone e contatos, conforme o nível de proteção da permissão e as configurações do usuário.

Permissões

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.INTERNET"/>
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>

    <application
```

Permissões

Permite que o aplicativo abra conexões de rede e acesse a Internet, como para consumir uma API REST.

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.INTERNET"/>
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>

    <application
```

Permite que o aplicativo consulte o estado da conexão de rede, verificando, por exemplo, se o dispositivo está conectado à Internet.

Retrofit

```
import retrofit2.Retrofit
import retrofit2.converter.gson.GsonConverterFactory
```

```
object RetrofitClient {

    2 Usages
    fun getInstance(baseUrl: String): Retrofit {
        return Retrofit.Builder()
            .baseUrl(baseUrl)
            .addConverterFactory(GsonConverterFactory.create())
            .build()
    }
}
```

Fast. Reliable. Automated.

A free CDN for open source projects

Optimized for JS and ESM delivery from npm and GitHub. Works with all web formats.

370 billion requests

past month

over 24 092 TB

past month

npm

ESM

GitHub

 **WordPress**

```
// load any project hosted on npm
https://cdn.jsdelivr.net/npm/package@version/file

// load jQuery v3.6.4
https://cdn.jsdelivr.net/npm/jquery@3.6.4/dist/jquery.min.js

// use a version range instead of a specific version
https://cdn.jsdelivr.net/npm/jquery@3.6/dist/jquery.min.js
https://cdn.jsdelivr.net/npm/jquery@3/dist/jquery.min.js

// omit the version completely to get the latest one
// you should NOT use this in production
https://cdn.jsdelivr.net/npm/jquery/dist/jquery.min.js

// add ".min" to any JS/CSS file to get a minified version
// if one doesn't exist, we'll generate it for you
https://cdn.jsdelivr.net/npm/jquery@3.6.4/src/core.min.js

// omit the file path to get the default file
https://cdn.jsdelivr.net/npm/jquery@3.6
```

README CC0-1.0 license

Free Currency Exchange Rates API



Publish-Currencies passing

Features:

- Free & Blazing Fast response
- No Rate limits
- 200+ Currencies, Including Common Cryptocurrencies & Metals
- Daily Updated

Contributors 2

-  **fawazahmed0** Fawaz Ahmed
-  **ZeN220**

Languages

Python 92.4% JavaScript 7.6%

<https://github.com/fawazahmed0/exchange-api#endpoints>

Endpoints:

- `/currencies`

Lists all the available currencies in prettified json format:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies.json>

Get a minified version of it:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies.min.json>

Retrofit

```
{  
  "linch": "linch",  
  "aave": "Aave",  
  "ada": "Cardano",  
  "aed": "Emirati Dirham",  
  "afn": "Afghan Afghani",  
  "akt": "Akash Network",  
  "algo": "Algorand",  
  "all": "Albanian Lek",  
  "amd": "Armenian Dram",  
  "amp": "Amp",  
  "ang": "Dutch Guilder",  
  "aoa": "Angolan Kwanza",  
  "ape": "ApeCoin",  
  "apt": "Aptos",  
  "ar": "Arweave",  
  "arb": "Arbitrum",  
  "ars": "Argentine Peso",  
  "atom": "Cosmos",  
  "ats": "Austrian Schilling",  
}
```

Endpoints:

- `/currencies`

Lists all the available currencies in prettified json format:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies.json>

Get a minified version of it:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies.min.json>

Retrofit

- `/currencies/{currencyCode}`

Get the currency list with EUR as base currency:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies/eur.json>

Get the currency list with EUR as base currency on date 2024-03-06:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@2024-03-06/v1/currencies/eur.json>

Get the currency list with BTC as base currency:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies/btc.json>

Get the currency list with BTC as base currency in minified format:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies/btc.min.json>

Retrofit

- `/currencies/{currencyCode}`

Get the currency list with EUR as base currency:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies/eur.json>

Get the currency list with EUR as base currency on date 2024-03-06:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@2024-03-06/v1/currencies/eur.json>

Get the currency list with BTC as base currency:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies/btc.json>

Get the currency list with BTC as base currency in minified format:

<https://cdn.jsdelivr.net/npm/@fawazahmed0/currency-api@latest/v1/currencies/btc.min.json>



```
{
  "date": "2026-10-02",
  "eur": {
    "linch": 10.70086755,
    "aave": 0.0060370059,
    "ada": 4.39676601,
    "aed": 4.13278675,
    "afn": 72.60750955,
    "akt": 1.68451396,
    "algo": 8.83162814,
    "all": 92.06357804,
    "amd": 410.74712474,
    "amp": 1776.48434765,
    "ang": 2.03382346,
    "aoa": 1033.57907917,
    "ape": 7.37421053,
    "apt": 1.35063379,
    "ar": 0.25683544,
    "arb": 5.40892782,
    "ars": 1715.69790267,
    "atom": 0.64171983,
    "ats": 13.7603,
```

Retrofit

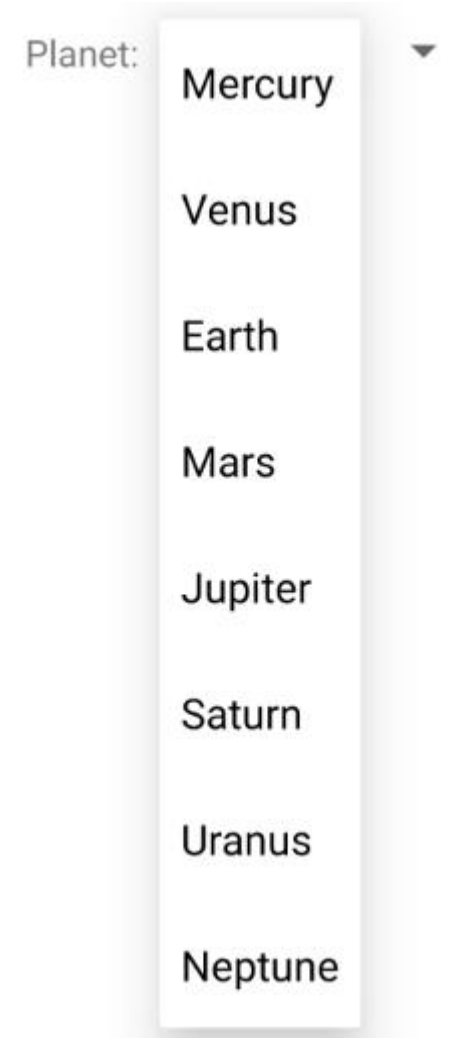
```
import com.google.gson.JsonObject
import retrofit2.Call
import retrofit2.http.GET
import retrofit2.http.Path
```

```
interface CurrencyApiService {
    1 Usage
    @GET(value = "npm/@fawazahmed0/currency-api@latest/v1/currencies.json")
    fun getCurrencies() : Call<JsonObject>

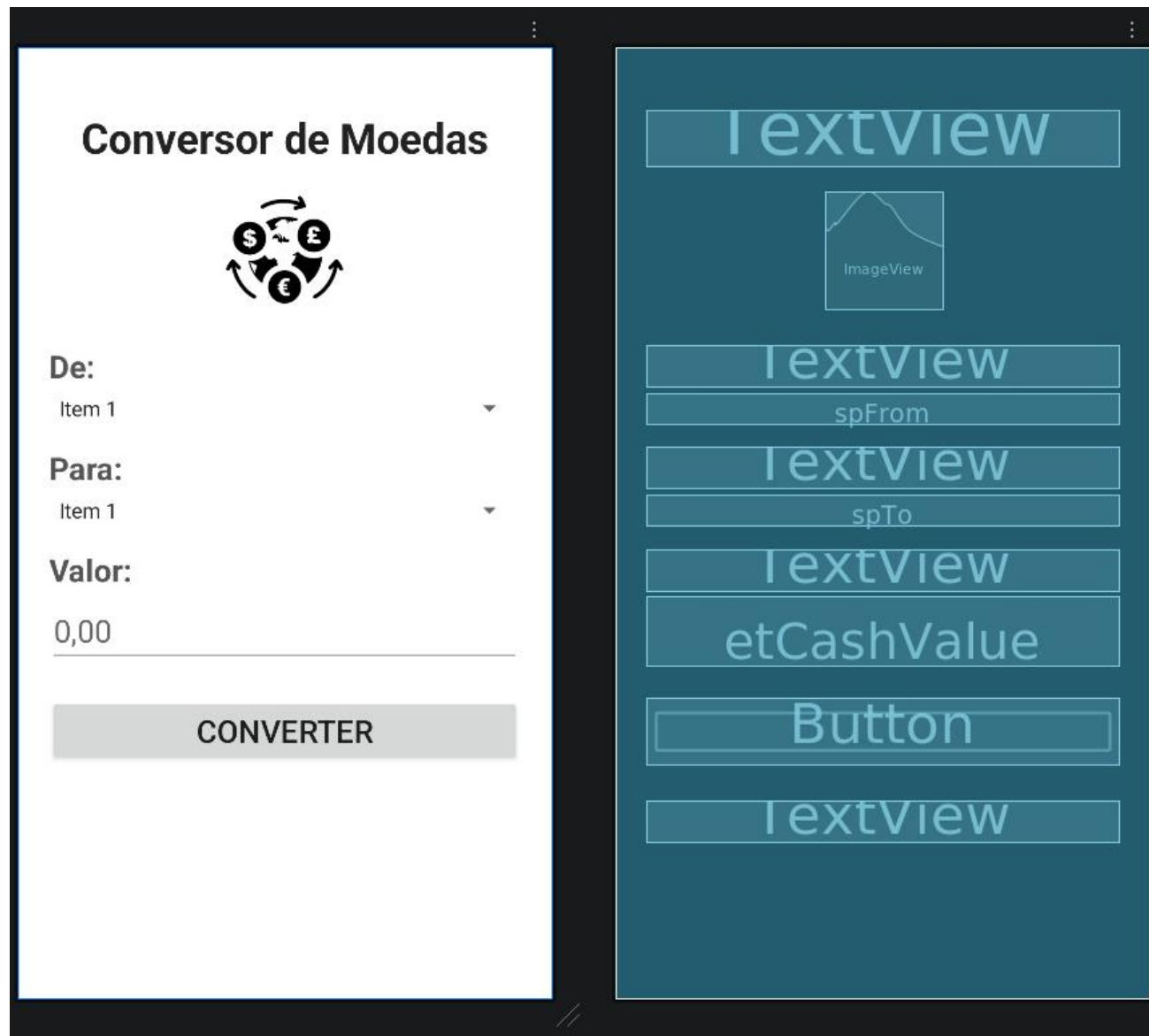
    1 Usage
    @GET(value = "npm/@fawazahmed0/currency-api@latest/v1/currencies/{from}.json")
    fun getCurrencyRate(
        @Path(value = "from", encoded = true) from : String
    ) : Call<JsonObject>
}
```

Spinner

- O Spinner é um componente de interface do usuário (UI) que permite ao usuário selecionar uma opção a partir de uma lista suspensa.
- Ao ser acionado, o Spinner apresenta as opções disponíveis para que o usuário selecione uma delas.
- Documentação:
 - <https://developer.android.com/develop/ui/views/components/spinner?hl=pt-br>



Retrofit



Retrofit

```
class MainActivity : ComponentActivity() {  
  
    3 Usages  
    private lateinit var binding: ActivityMainBinding  
  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        binding = ActivityMainBinding.inflate(layoutInflater)  
        setContentView(binding.root)  
  
        getCurrencies()  
  
        binding.btnConvert.setOnClickListener {  
            convertCurrency()  
        }  
    }  
}
```

Retrofit

```
private fun getCurrencies() {  
  
    val retrofitClient = RetrofitClient.getInstance(baseUrl = "https://cdn.jsdelivr.net/")  
  
    val service = retrofitClient.create(CurrencyApiService::class.java)  
  
    service.getCurrencies().enqueue(callback = object : Callback<JsonObject> {  
  
        override fun onResponse(call: Call<JsonObject>, response: Response<JsonObject>) {  
  
            if (!response.isSuccessful) {  
                showToast("Erro HTTP: ${response.code()}")  
                return  
            }  
  
        }  
  
    })  
}
```



Retrofit

```
val body = response.body()

if (body == null) {
    showToast("Sem resposta da API")
    return
}

val data = body.keySet().toList()

val adapter = ArrayAdapter(context = this@MainActivity,
    resource = android.R.layout.simple_spinner_item, objects = data)

adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item)

binding.spFrom.adapter = adapter
binding.spTo.adapter = adapter
```

Retrofit



```
val brlPosition = data.indexOf("brl")
val usdPosition = data.indexOf("usd")

if (brlPosition >= 0) {
    binding.spFrom.setSelection(brlPosition)
}

if (usdPosition >= 0) {
    binding.spTo.setSelection(usdPosition)
}
}

override fun onFailure(call: Call<JsonObject>, t: Throwable) {
    showToast("Erro: ${t.message}")
}

})
}
```

Retrofit

```
fun showToast(message: String){  
    Toast.makeText(context = this, text = message, duration = Toast.LENGTH_LONG).show()  
}
```

Retrofit

```
private fun convertCurrency() {  
  
    val from = binding.spFrom.selectedItem.toString()  
    val to = binding.spTo.selectedItem.toString()  
  
    val value = binding.etCashValue.text.toString().toDoubleOrNull()  
  
    if (value == null) {  
        showToast("Informe um valor válido")  
        return  
    }  
  
    val retrofitClient = RetrofitClient.getInstance(baseUrl = "https://cdn.jsdelivr.net/")  
  
    val service = retrofitClient.create(CurrencyApiService::class.java)
```



Retrofit

```
service.getCurrencyRate(from).enqueue(p0 = object : Callback<JsonObject> {  
  
    override fun onResponse(call: Call<JsonObject>, response: Response<JsonObject>) {  
  
        if (!response.isSuccessful) {  
            showToast("Erro HTTP: ${response.code()}")  
            return  
        }  
  
        val body = response.body()  
  
        if (body == null) {  
            showToast("Resposta vazia da API")  
            return  
        }  
    }  
})
```

Retrofit

```
val currencyData = body.getAsJsonObject(memberName = from)

if (currencyData == null) {
    showToast("Moeda de origem não encontrada")
    return
}

val rate = currencyData.get(to)?.asDouble

if (rate == null) {
    showToast("Taxa de conversão não encontrada")
    return
}

val conversion = value * rate

val result = "${to.uppercase()} $conversion"
```

Retrofit

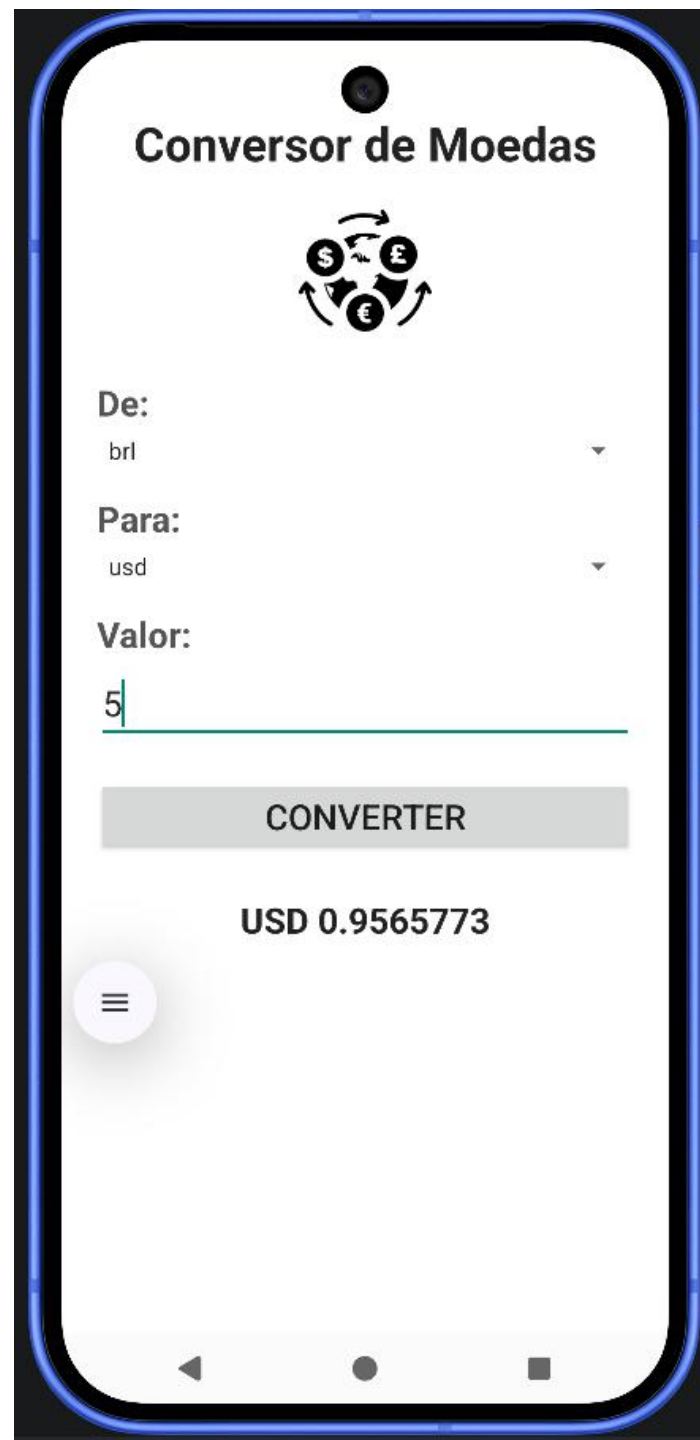


```
        binding.tvResult.text = result
    }

    override fun onFailure(call: Call<JsonObject>, t: Throwable) {
        showToast("Erro: `${t.message}`")
    }
}

)
```

Retrofit



Exercícios

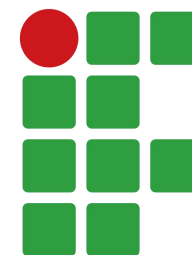
- Utilize uma API desenvolvida em outra disciplina e implemente, um aplicativo Android, com as operações de comunicação com a API utilizando os métodos HTTP GET, POST, PUT/PATCH e DELETE, por meio das anotações @GET, @POST, @PUT/@PATCH e @DELETE do Retrofit.
 - Caso não possua uma API própria, utilize a API pública DummyJSON (<https://dummyjson.com/>).

Dúvidas



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